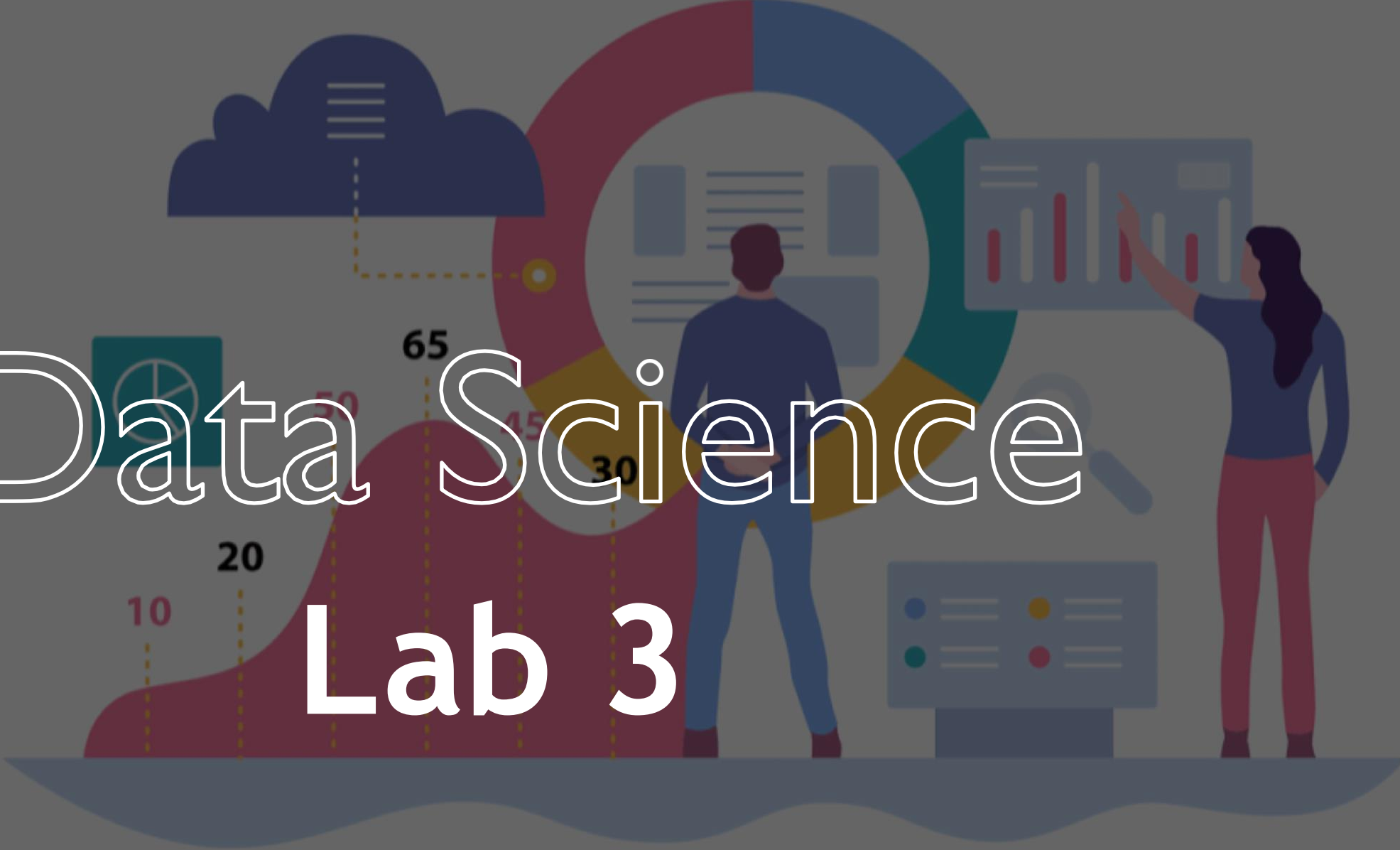


# Data Science

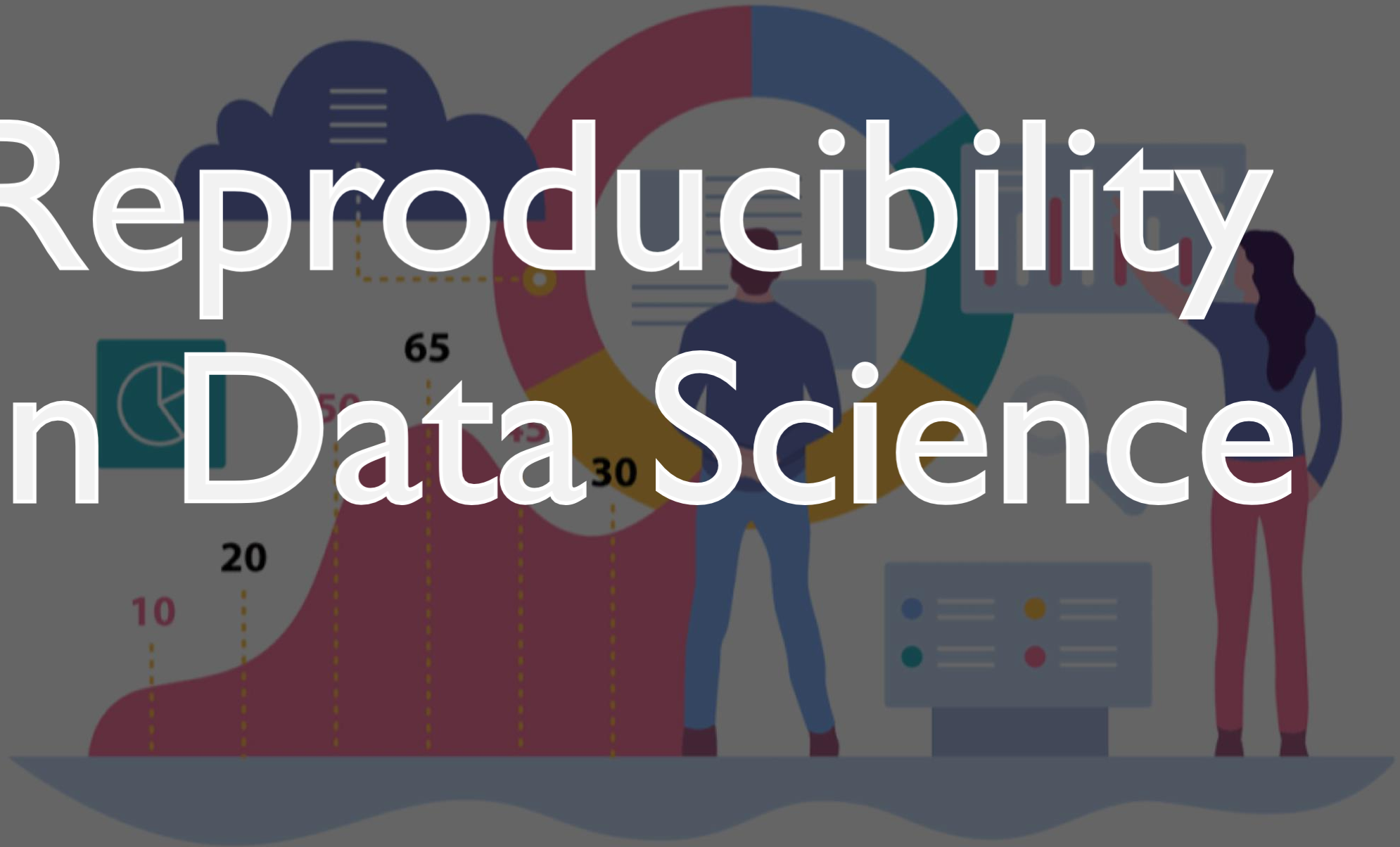
A stylized illustration in a flat, modern style. In the center, a man in a blue shirt and blue pants stands with his back to the viewer, looking at a large circular dashboard with various charts and data points. To his right, a woman in a blue shirt and red pants stands, pointing at a bar chart on a screen. The background is filled with various data-related icons: a cloud with a list, a pie chart, a bar chart, a line graph with numbers (10, 20, 30, 45, 50, 65), a magnifying glass, and a computer monitor displaying a dashboard. The overall color palette is muted, with blues, greys, and earthy tones.

## Lab 3

# Objectives

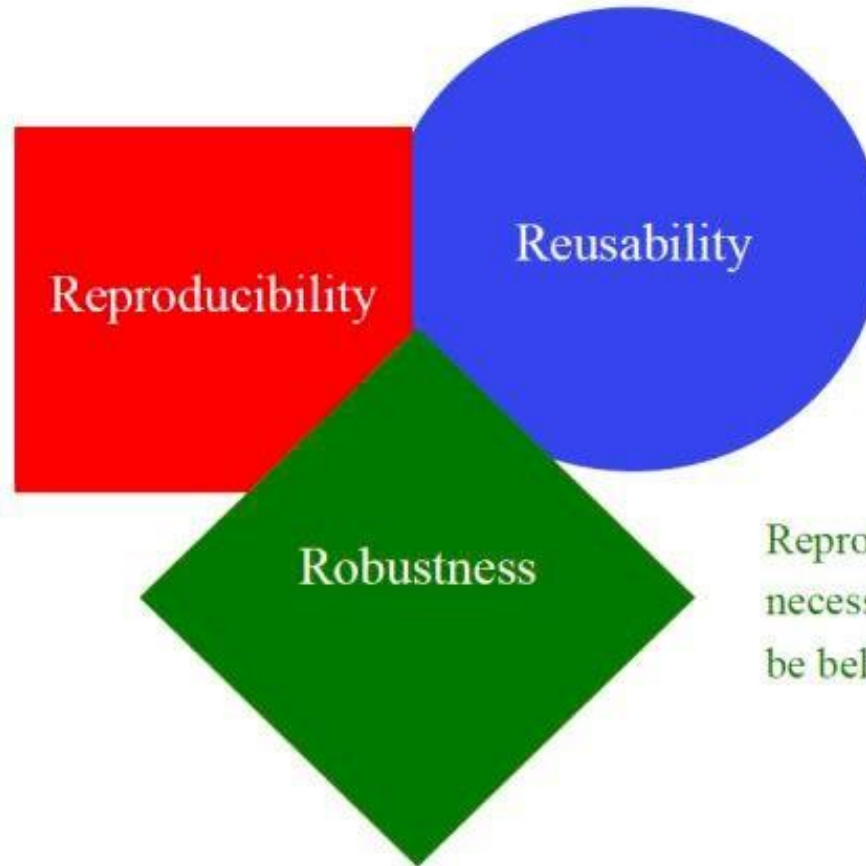
- Understand ongoing research in the field of data science
- Identify research claims from papers
- Reproduce results using code/data provided by the authors

# Reproducibility in Data Science



# Reproducibility in Data Science

**“Reproducibility** refers to the ability of a researcher to duplicate the results of a prior study....

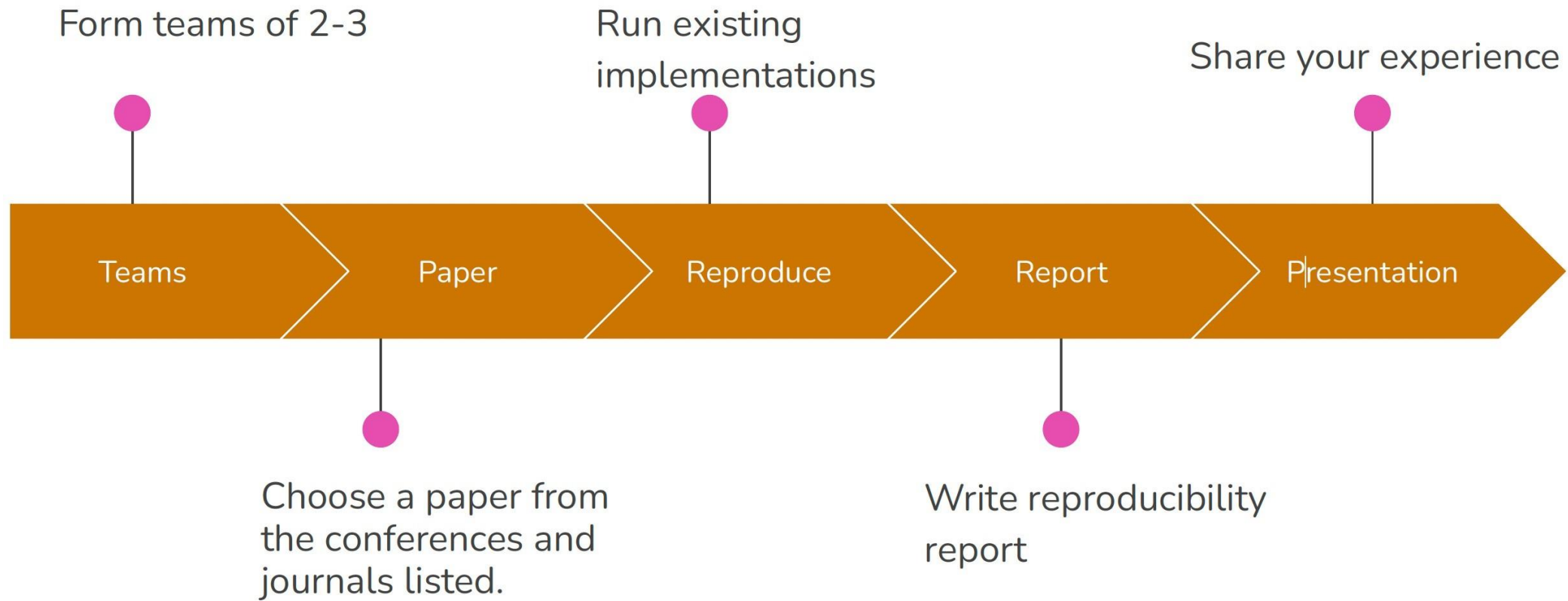


Using the same materials as were used by the original investigator.

Reproducibility is a minimum necessary condition for a finding to be believable and informative.”

Bollen et al.  
National Science Foundation, 2015.

# Reproducibility in Data Science



# Choosing a Paper

- Produce reproducibility reports on papers from:
  - Top conferences in AI/ML including: [NeurIPS](#), [ICML](#), [ICLR](#), [ACL](#), [EMNLP](#), [CVPR](#), [AAAI](#), [IJCAI](#), [ACM FAccT](#), [SIGIR](#), [ECCV](#).
  - Top ML journals, including: [JMLR](#), [TACL](#) and [TMLR](#).

# Reproduce Results

- Identify the main claims of the paper.
- Run the code provided by authors to validate the claims.
- Contact authors.
- Tune hyperparameters.

# Reproducibility Report & Presentation

- Template: will be provided.
- Maximum Length: 9 pages.
- GitHub Repository with the code.
- Discuss encountered challenges, along with recommendation.
- Deadline: **06.11.2024**.